Available online at www.sciencedirect.com
ScienceDirect
Comput. Methods Appl. Mech. Engrg. 416 (2023) 116384
Computer methods in applied mechanics and engineering
www.elsevier.com/locate/cma

Solving nonconvex energy minimization problems in martensitic phase transitions with a mesh-free deep learning approach

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Received 5 June 2023; received in revised form 11 August 2023; accepted 15 August 2023
Available online xxxx

Abstract

In this paper, we propose a novel mesh-free method for solving nonconvex energy minimization problems for martensitic phase transitions and twinning in crystals using the deep learning approach. These problems pose significant challenges to analysis and computation because they involve multiwell gradient energies with large numbers of local minima, each involving a topologically complex microstructure of free boundaries with gradient jumps. We use the Deep Ritz method, which represents candidates for minimizers using parameter-dependent deep neural networks, and minimize the energy with respect to network parameters. The key feature of our method is a newly proposed activation function called SmReLU, which captures the structure of minimizers where traditional activation functions fail. Our mesh-free approach allows for the approximation of free boundaries, essential to this problem, without any special treatment, making it extremely simple to implement. We demonstrate the success of our method through numerous numerical computations.

AMS subject classifications: 35R05; 65N30; 68T99; 74A50; 74G65

Keywords: Deep learning method; Variational problems; Mesh-free method; Phase transition problems; Nonconvex energy minimization

1. Introduction

Physics-informed deep learning methods [1–10] have recently achieved great success in solving partial differential equation (PDE) problems arising in different fields of STEM. This entails using deep neural networks to approximate the solution of PDEs, often in many dimensions, leading to a mesh-free numerical method in place of more established schemes, such as finite element methods or finite difference methods.

In this approach [8,11], the loss function to be minimized is the mean square of the PDE residual. This approach is widely known to converge for linear elliptic and parabolic problems [10], and it has been successful in solving a variety of nonlinear PDEs [12,13]. However, it is unclear how well this approach would perform in situations where the PDE has weak solutions with reduced smoothness. Examples of such situations include hyperbolic

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https://doi.org/10.1016/j.cma.2023.116384
0045-7825/

conservation laws in nonlinear elastodynamics, and solid–solid phase transitions described by nonconvex gradient energy functions in nonlinear elastostatics [14–16]. In the latter case, the associated Euler–Lagrange equations change type and lose ellipticity at certain values of the solution gradient [15]. In these situations, there are weak solutions with discontinuous gradients across interfaces that are unknown *a priori* but are part of the solution.

Another type of interface problem involves heterogeneous media that have high-contrast or discontinuous features specified *a priori*. Examples include PDEs whose coefficients are discontinuous across a fixed interface, resulting in a loss of smoothness in solutions. To address these problems, various sophisticated finite element methods (FEM) and finite difference methods (FDM) have been developed [17–23]. While these methods are highly accurate, they often require special treatment of the interface and jump conditions across it when creating the finite element mesh and associated basis functions for intersecting elements.

Recently, deep learning was used to solve interface problems involving linear elliptic PDE systems with discontinuous coefficients and/or discontinuous forcing [24], motivated by biomechanics [25]. The problems studied in [24] are solved by energy minimization. The energy functional is strictly convex, and a weak solution of the PDE (Euler–Lagrange equation) is its unique minimizer. The interface is fixed at the location of the coefficient jump in the PDE. The Deep Ritz method [5] used in [24] approximates solutions by a deep neural network (DNN) and minimizes the energy as a function of the DNN parameters (weights and biases). We also refer the interested reader to [26] for a variational neural network approach for glacier modeling with nonlinear rheology. The DNN method is mesh-free and does not require any special structure to capture the location of the interface inside the domain spontaneously and accurately. Another important result [27] is the use of the DNN-based deep learning method to investigate the relation between rank-one convexity and quasiconvexity, known as Morrey's Conjecture in the calculus of variations, in the context of nonlinear elasticity.

In phase transition problems, interfaces that separate different phases of a body (e.g., solid and liquid or austenite and martensite) are free boundaries whose position or shape is unknown *a priori*. Instead, they are part of the solution and can evolve into time-dependent problems. Phase boundaries (e.g., domain walls, twin boundaries) in different physical settings are characterized by a jump discontinuity or narrow transition layer of the unknown scalar or vector field (e.g., order parameter, deformation, polarization) or its spatial derivatives. To identify the unknown phase boundary, special augmented models have been developed, such as phase field and level set methods. The Allen–Cahn and Cahn–Hilliard equations are prototypical phase field models that typically involve a nonconvex, multi-well energy density function. The DNN approach has been used to solve free boundary problems of the Stefan type [28] and to describe phase boundaries in the Allen–Cahn and Cahn–Hilliard equations [29,30]. The method utilizes standard DNNs without requiring any modification to identify phase boundaries.

Martensitic phase transitions [14–16] are frequently modeled using a free energy density that is a nonconvex, multi-well potential dependent on the *gradient* of the unknown deformation field. In fact, the martensitic phase itself involves two or more symmetry-related variants, known as twins, that correspond to different energy wells.

The fact that the energy depends on the gradient of the unknown function (deformation) introduces special challenges. Local energy minimizers involve phase boundaries that separate different phases of a body. Across these interfaces, the deformation field remains continuous, but its gradient jumps. The continuity of the deformation field imposes restrictions on the jump of its gradient (Hadamard compatibility conditions [15,16]). In many crystalline alloys, the austenite and martensite energy wells violate these conditions, which are nonetheless satisfied between martensitic twin variants. As a result, no absolute energy minimum can be achieved [16]. Instead, the energy approaches its infimum in the limit, along a sequence of deformations with a laminated microstructure of parallel twin interfaces. These minimizing sequences involve an increasing number of twin boundaries and converge to a limit state compatible with austenite [16]. This microstructure is known as finely twinned martensite, and also arises from incompatibility between twins of different orientations, as observed in experiments, Fig. 1(a), and/or rigid (null Dirichlet) boundary conditions, Fig. 1(b, c). Such laminated microstructures were observed in experiments, as shown in Fig. 1(a).

Near an incompatible boundary, twin layers can taper into needles, and it is often observed that individual layers split into two or more needles in experiments (Fig. 1(a)) [31,33]. This phenomenon, known as twin branching, has been extensively studied both analytically [32,34–36] and numerically [32,37–39] as a mechanism for achieving an energy minimum in the presence of interfacial energy. To capture this complex behavior, it is possible to consider some lower-dimensional problems. Many of these problems have an energy of the form as follows:

$$\int_{\Omega} W(\nabla u(\boldsymbol{x}))d\boldsymbol{x}, \tag{1}$$

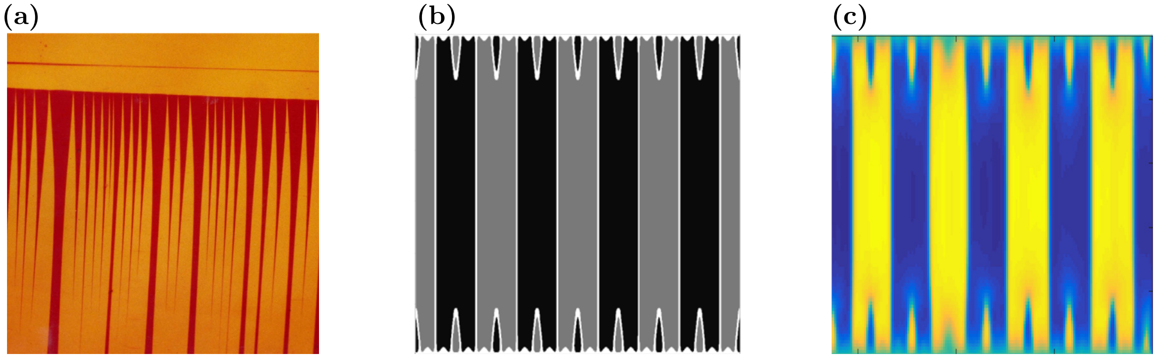

Fig. 1. Twinned Microstructures: (a) Experimental in CuAlNi single crystal [31](reproduced here with permission of the authors). **(b), (c):** Comparison of results of different numerical methods for a simplified 2D model: (b) Finite Difference solution [32] (reproduced here with permission of the authors) and (c) our DNN solution of the 2D regularized problem (10).

where $\Omega \subset \mathbb{R}^n$, $u : \Omega \to \mathbb{R}^m$ and $n = 2$ or 3, while $m = n$ or $m = 1$ for the vector and scalar case, respectively. In all cases, $W : \mathbb{R}^{m\times n} \to \mathbb{R}$ is a nonconvex function with multiple wells. Letting the derivative of W be $S : \mathbb{R}^{m\times n} \to \mathbb{R}^{m\times n}$, the Euler–Lagrange equation of (1) is

$$\operatorname{div} S(\nabla u) = 0 \quad \text{in } \Omega. \tag{2}$$

Because of loss of rank one convexity [15,16] for values of ∇u between wells, this PDE system loses ellipticity. Because of ellipticity loss, (2) has weak solutions with phase boundaries, across which ∇u jumps discontinuously from phase to phase [15] and second derivatives fail to exist. This precludes using the physics-informed neural network method (PINN), which requires the strong form of (2) to be valid [7]. An added problem is the extensive loss of uniqueness of weak solutions of (2) [14], many of which are energetically unstable (not even local energy minima). The alternative is to minimize the energy (1) [16,34–36,39]. We use the Deep Ritz method [4], namely energy minimization in terms of DNNs. An additional advantage is that piecewise smooth local minimizers of (1) are also weak solutions of (2) which are at least metastable, in contrast to some solutions of (2).

In this study, we investigate the application of the Deep Ritz method to energy minimization in the context of nonlinear elasticity. Specifically, we apply this method to a range of one and two-dimensional problems involving nonconvex potentials of the deformation gradient. Our results demonstrate that the Deep Ritz method is capable of spontaneously capturing complex microstructures, such as finely twinned structures that exhibit gradient jumps across multiple curved interfaces and splitting/tapering topological transitions near boundaries [32–36,38,39]. Notably, the method is mesh-free, which enables it to accurately describe curved interfaces of arbitrary orientation. This is in contrast to FEM and other mesh-dependent methods [37]. Importantly, the Deep Ritz method does not require any special structure of the deep neural network used, except for a specially designed activation function that is suggested by the structure of solutions to simple problems.

We find that the activation function plays a crucial role in our proposed method. Our first choice is guided by the observation that in the simplest one-dimensional problem, an exact solution is provided by the piecewise linear ReLU activation function. We then extend our approach to consider the regularization of the problem by including the second derivative of the deformation in the energy [40]. Minimizers are smooth, with transition layers replacing derivative jumps. To capture these smooth minimizers, we introduce a modified version of the ReLU activation function, which we call SmReLU. In two-dimensional (2D) problems, the use of SmReLU allows us to capture spatially complex microstructures with fine twinning and twin branching, Fig. 1(c), whereas traditional activation functions fail to do so.

The outline of this paper is as follows: In Section 2, we introduce the Deep Ritz method for nonconvex energy minimization in solving martensitic phase transitions. Details about the implementation of the deep neural network, including the choices of activation function and how to impose boundary conditions will be presented in this section. In Section 3, we present numerical results to demonstrate the performance of our method. Finally, concluding remarks are made in Section 4.

2. Methods

2.1. The minimization problem

The problem we study in this paper concerns the attempt to minimize a nonconvex functional that is a low-dimensional model for an austenite–martensite phase transition. The problem involves finding

$$\min E\{u\}, \quad E\{u\} = \int_\Omega W(\nabla u(\boldsymbol{x}))d\boldsymbol{x} \tag{3}$$

over suitable functions $u : \Omega \to \mathbb{R}$, where the function $W : \mathbb{R}^2 \to \mathbb{R}$ is a double-well potential. Here the domain $\Omega = [0, L] \times [0, 1] \subset \mathbb{R}^2$ and we choose $W(\nabla u) = \frac{1}{2}\left(u_x^2(u_x - 1)^2 + u_y^2\right)$ with $\nabla u = (u_x, u_y)$ [34,35]. The wells (minima) of W are at $(0, 0)$ and $(1, 0)$. Moreover, u in (3) is subject to the Dirichlet boundary conditions

$$u(x, y) = \gamma x \quad \forall\, (x, y) \in \partial\Omega. \tag{4}$$

Of particular interest is the case $\gamma = 1/2$ as the linear function $u(x, y) = x/2$ for all $(x, y) \in \Omega$ satisfying the boundary conditions has $\nabla u = (1/2, 0)$, which is a saddle point of the function W. The zero-energy minima $u = 0$ and $u = x$ on Ω are incompatible with the boundary conditions, whereas there are sequences of functions approaching zero energy, namely problem (3) does not have a minimizer but only minimizing sequences. This means one can construct sequences of functions u_n with $E\{u_n\} \to 0$ as $n \to \infty$. This occurs due to geometric incompatibility of the energy-minimal states $u = \text{const}$ and $u = x + \text{const}$ on Ω with the boundary conditions (4).

On the other hand, the mixed boundary conditions

$$u(0, y) = 0, \quad u(1, y) = \gamma, \quad 0 < y < 1, \tag{5}$$

at the vertical sides $x = 0, L$ with the horizontal sides $y = 0, 1$ free, are compatible with the ansatz $u(x, y) = u(x)$, $0 < y < 1$ (using u for both functions by notation abuse) which reduces (3) to the following one-dimensional (1D) problem. Let $\Omega = [0, 1]$. We define:

$$W(z) = z^2(1 - z)^2, \quad z \in \mathbb{R},$$

and one can see that $W(z)$ is a double-well potential with minima at 0 and 1. Suppose $u : [0, 1] \to \mathbb{R}$ satisfies $u(0) = 0$ and $u(1) = \gamma$ for a given constant $\gamma \in \mathbb{R}$. Then, the minimization problem (3) reduces to minimize

$$\min \int_0^1 W(u'(x))dx, \tag{6}$$

where u' is the derivative of u. We define

$$f(x) = \frac{x + |x|}{2}, \quad x \in \mathbb{R}. \tag{7}$$

Then, if $0 < \gamma < 1$, a solution of the 1D problem (6) (with zero energy) is

$$u(x) = f\left(x - 1 + \gamma\right). \tag{8}$$

This is piecewise smooth with a derivative discontinuity at $x = 1 - \gamma$.

Remark 2.1. Any partition of $[0, 1]$ into intervals where the slope of u alternates between 0 and 1 (while u remains continuous) is a minimizer of the 1D problem (6) with zero energy. This illustrates the massive loss of uniqueness of minimizers of the 1D problem (6), but also the 2D problem with boundary conditions (5).

Remark 2.2. In order to make the problem (6) well-posed, it is common to add a higher gradient regularization, also known as capillarity, to the energy $E\{u\}$, which now becomes

$$\min E_\varepsilon\{u\}, \quad E_\varepsilon\{u\} = \int_0^1 \left(W(u'(x))dx + \frac{\varepsilon^2}{2}\left(u''(x)\right)^2\right) dx, \tag{9}$$

where $\varepsilon > 0$ is the higher-gradient coefficient, a small parameter [40]. It is known that (9) has an essentially unique solution (modulo reflection about $x = 1/2$) [40], where the gradient discontinuity of (8) is replaced by a single, smooth transition layer between values 0 and 1 of the derivative u'. For small ε the thickness of this layer is of order

ε and the gradient term contributes an interfacial or surface energy to the interface, also of order ε. The analogous 2D regularized energy that we also study in this work is

$$E_\varepsilon\{u\} = \int_\Omega \left(W(\nabla u(x,y)) + \frac{\varepsilon^2}{2}\Big[u_{xx}(x,y)\Big]^2 \right) dxdy. \tag{10}$$

This is a much more complex energy functional with multiple local minima [32], depending on the values of ε and boundary conditions. Here a global minimizer does exist under suitable growth conditions on W, but uniqueness is not known, and the multitude of local minima is studied in [32].

2.2. Definition of deep neural networks (DNNs)

To define a deep neural network (DNN), we need two ingredients. The first is a (vector) linear function $T : \mathbb{R}^n \to \mathbb{R}^m$ of the form $T(\mathbf{x}) = A\mathbf{x} + b$, where $A = (a_{ij}) \in \mathbb{R}^{m\times n}$, $\mathbf{x} \in \mathbb{R}^n$, and $b \in \mathbb{R}^m$. The second is a nonlinear activation function $\sigma : \mathbb{R} \to \mathbb{R}$. While the sigmoid function is a popular choice in the artificial neural network literature, we will use another activation function, known as the rectified linear unit (ReLU), defined by $\sigma(x) = \max(0, x)$ [41]. Note that $\sigma(x) = f(x)$ defined in (7), which provides a solution of the 1D minimization problem (6) in (8). By applying the activation function to each component x_j $(j = 1,\ldots,n)$ of the vector $x \in \mathbb{R}^n$, we can define a vector activation function $\sigma : \mathbb{R}^n \to \mathbb{R}^n$ with components $\sigma(x_j)$, $j = 1,\ldots,n$.

Equipped with those definitions, we are able to construct a continuous function $F(\mathbf{x})$ as an alternating composition of linear transformations and (vector) activation functions as follows:

$$F(\mathbf{x}) = T^k \circ \sigma \circ T^{k-1} \circ \sigma \cdots \circ T^1 \circ \sigma \circ T^0(\mathbf{x}). \tag{11}$$

Here $T^i(\mathbf{x}) = W_i\mathbf{x}+b_i$ is an affine transformation, with the weights W_i as of yet unspecified matrices and the biases b_i unspecified vectors. Also $\sigma(\cdot)$ is the component-wise defined activation function. Dimensions of W_i and b_i are chosen to make (11) meaningful. The parametrization (11) is called a $(k+1)$-layer DNN, which has k hidden layers. Denoting all the undetermined coefficients (namely W_i and b_i) in (11) as $\theta \in \Theta$, where θ is a high-dimensional vector and Θ is the space of θ. The DNN representation of a continuous function can be viewed as

$$F = F(\mathbf{x}; \theta). \tag{12}$$

Let $\mathbb{F} = \{F(\cdot,\theta)|\theta \in \Theta\}$ denote the set of all functions expressible by DNNs parametrized by $\theta \in \Theta$ as in (11). Then $\mathbb{F}$ provides an efficient way to represent unknown continuous functions [2], compared to a linear solution space used in classic numerical methods, e.g., a trial space spanned by linear nodal basis functions in the FEM.

Remark 2.3. In the computation, we propose a Smoothened Rectified Linear Unit (SmReLU) activation function to approximate u for the regularized energy problems (9) and (10). We refer readers to Section 3 for details.

2.3. The DNN representation of the nonconvex energy minimization

Given the energy functional $E\{u\}$ from (3), we can derive a DNN-based numerical method to solve the minimization problem in order to compute the solution u, in the spirit of the Deep Ritz method [5]. Here we seek the solution of the energy minimization problem

$$u = \operatorname*{argmin}_{v\in\mathbb{H}^1(\Omega)} E\{v\}, \tag{13}$$

subject to the specified boundary condition $u(x) = g(x)$ on $\partial\Omega$.

From the perspective of scientific computing, the energy minimization problem (13) can be solved using numerical methods, such as the FEM and FDM. From the perspective of machine learning, however, the numerical solution of $u(x)$ is interpreted as a function with $\mathbf{x} \in \mathbb{R}^2$ as its input and $u \in \mathbb{R}^1$ as its output, which can be approximated by a DNN $F(\mathbf{x})$ defined in (11). As a result, problem (13) is replaced by the problem

$$\tilde{u} = \operatorname*{argmin}_{F\in\mathbb{F}_b} E\{F\}, \tag{14}$$

Here $\tilde{u}$ denotes the DNN solution to the problem of minimizing the energy over the DNNs (11) in $\mathbb{F}$ and $\mathbb{F}_b$ denote the set of all functions expressible by DNNs satisfying the boundary conditions. Let

$$J(\theta) = E\{F(\cdot, \theta)\} = \int_{\Omega} W(\nabla F(x, \theta))dx, \quad \theta \in \Theta, \tag{15}$$

where F is the DNN solution. Then, the minimization problem (14) reduces to a finite-dimensional optimization problem

$$\tilde{\theta} = \underset{\theta \in \Theta}{\operatorname{argmin}}\, J(\theta), \quad \tilde{u}(x) = F(x, \tilde{\theta}). \tag{16}$$

This yields $\tilde{u}$, the DNN representation of the solution of (14).

Notice that the original energy minimization problem (13) is non-convex. The variational problem (16) is non-convex as well. Clearly, the issue of local minima and saddle points is nontrivial, which brings essential challenges to many existing optimization methods. Since the parameter space Θ is typically very large, one often uses the stochastic gradient descent (SGD) method [42] to solve (16).

How to impose boundary conditions in the DNN method is an important issue [11]. For the Dirichlet boundary condition, we use the equivalent of the Nitsche method in FEM for DNN, known as the Deep Nitsche Method [43]. We add a boundary penalty term in the energy function to address this issue. Specifically, one adds a soft constraint (i.e., a boundary integral term) to the energy $J(\cdot)$ defined in (15) and obtains

$$\tilde{u}_\tau = \underset{\theta \in \Theta}{\operatorname{argmin}}\Big[J(\theta) + \tau \int_{\partial\Omega} \big(F(x, \theta) - g(x)\big)^2 dx\Big], \tag{17}$$

where $\tau > 0$ is the penalty parameter to scale and balance the losses from the energy part and the boundary part. The idea is that the term $\int_{\partial\Omega} \big(F(x,\theta) - g(x)\big)^2 dx$ will approach zero when we increase the value of τ in the calculation, whereas the Dirichlet boundary condition $u = g$ on $\partial\Omega$ will be satisfied in a weak L^2 sense, and one hopes that $\tilde{u}_\tau \to \tilde{u}$ as $\tau \to \infty$.

It has been recently pointed out that the penalty parameter τ should not be chosen arbitrarily but adaptively in a rational way stemming from the specific problem [44]. Here we also point out that the parameter τ can be given physical meaning as the stiffness of the austenite phase that exists beyond the boundary $y = 1$, whereas the martensite phase occupies Ω. The exact satisfaction of boundary conditions such as (4) would correspond to infinitely rigid austenite.

2.4. Implementation of the SGD method

Since the number $\dim \Theta$ of degrees of freedom in the optimization problem (16) is quite large, we apply the SGD method on the parameter space Θ to solve it. Notice that θ is a high-dimensional vector and let θ_k be any component of θ. We approximate the corresponding component $\partial J(\theta)/\partial \theta_k$ of the gradient of $J(\theta)$ as follows:

$$\begin{aligned} \frac{\partial J(\theta)}{\partial \theta_k} &= \int_{\Omega} \frac{\partial\big(W(\nabla F(x,\theta))\big)}{\partial \theta_k} dx + \tau \int_{\partial\Omega} \frac{\partial\big(F(x,\theta) - g(x)\big)^2}{\partial \theta_k} ds \\ &\approx \frac{vol(\Omega)}{N} \sum_{i=i}^{N} \frac{\partial\big(W(\nabla F(x_i,\theta))\big)}{\partial \theta_k} + \frac{\tau S(\partial\Omega)}{N_b} \frac{\partial\big((F(y_j,\theta) - g(y_j))^2\big)}{\partial \theta_k}, \end{aligned} \tag{18}$$

where the collocation points $x_i \overset{i.i.d.}{\sim} Unif(\Omega)$ are uniform random points that are sampled from the physical domain Ω, $y_j \overset{i.i.d.}{\sim} Unif(\partial\Omega)$ are uniform random points that are sampled from the boundary $\partial\Omega$, $vol(\Omega)$ is the volume of the domain, and $S(\partial\Omega)$ is the perimeter of the domain. In the context of the deep learning method, N and N_b are called batch numbers, which means the number of training samples utilized in one iteration.

After we compute the approximation of the gradient with respect to θ_k, we can update each component of θ as

$$\theta_k^{n+1} = \theta_k^n - \eta \frac{\partial J(F(\cdot, \theta))}{\partial \theta_k}\Big|_{\theta_k = \theta_k^n}, \tag{19}$$

where η is the learning rate. In our numerical computation, we use the Adam optimizer [45], a modified version of the SGD method. Our results indicate that the Adam optimizer is very effective in solving this non-convex optimization problem.

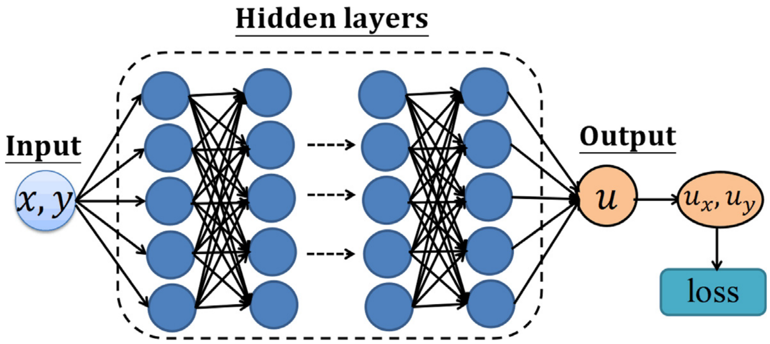

Fig. 2. Schematic of a DNN for 2D problems..

Remark 2.4. From the derivation of the DNN formulation, one can see that the proposed method automatically deals with the phase boundary (or discontinuity in the gradient of the solution) without knowing the locations of the discontinuity *a priori*. Thus, the proposed method is advantageous in dealing with phase transition problems with complex geometry.

Fig. 2 shows a possible network layout for approximating u with input (x, y). For the output u, we can use automatic differentiation techniques to compute derivatives of u that appear in the energy in the minimization problem.

3. Numerical results

In this section, we present numerical examples to demonstrate the performance of our proposed deep learning method in solving nonconvex energy minimization problems for martensitic phase transitions. We use a fully connected DNN (e.g. Fig. 2 for 2D problems) to carry out our computations. The weights are initialized with truncated normal distributions, with variances set as two over the sum of input and output unit numbers. The biases are initialized as zero. We use the Adam optimizer.

3.1. Numerical results for 1D problems

We first demonstrate how to use DNNs to minimize the energy in a 1D problem as follows:

$$\min \int_0^1 W(u'(x))dx, \tag{20}$$

where $W(z) = z^2(1-z)^2$, $z \in \mathbb{R}$ with $u(0) = 0$ and $u(1) = \gamma$. The integrand W has two wells at 0 and 1 and we choose typically choose $\gamma = 0.25, 0.5$ and 0.75. The solution u is approximated by a DNN $u_{NN}(x) = F(x; \theta)$, where x is the input of the DNN and θ is the vector of weights w_i and biases b_i as in (11), which are the trainable parameters in the DNN.

On one hand, the parameters θ of the DNN $u_{NN}(x)$ should minimize the energy in (20) (Deep Ritz Method [5]). We construct the corresponding loss function as follows:

$$loss_e(\theta) = \frac{1}{N}\sum_{i=1}^{N} W\left(\frac{\partial F}{\partial x}(x_i, \theta)\right), \tag{21}$$

where $\{x_i\}_{i=1}^N$ are the collocation points in Ω. The sum here approximates the integral in (20). The derivative $u'_{NN} = \partial F/\partial x$ is computed with automatic differentiation.

On the other hand, $u_{NN}(x)$ should satisfy the boundary conditions, which corresponds to minimizing the loss function

$$loss_b(\theta) = (F(0, \theta) - 0)^2 + (F(1, \theta) - \gamma)^2. \tag{22}$$

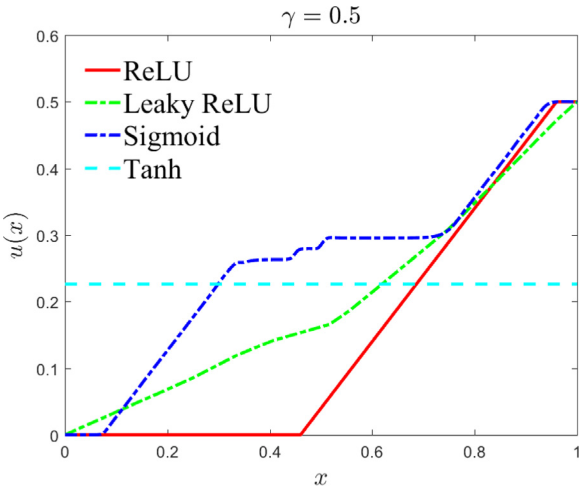

Fig. 3. The solution $u(x)$ of the problem (20) obtained by DNN methods with different activation functions and $\eta = 10^{-2}$.

This corresponds to the Deep Nitsche method [43]. Combining the two loss functions (21) and (22), the total loss function to be minimized is

$$loss(\theta) = loss_e(\theta) + \tau loss_b(\theta), \tag{23}$$

where τ is the penalty parameter to scale and balance the losses from the energy part and the boundary part. During training, the shared parameters θ are adjusted by back-propagating the error obtained by minimizing a loss function (23) that is the weighted sum of the above two constraints.

3.1.1. Activation function

In this problem, the activation function plays an important role. As observed in Section 2.1, an exact solution of (20) is provided by (8). We observe that in (8), the function $f(x)$ defined by (7) is identical with the well-known ReLU activation function

$$\text{ReLU:}\ \ \sigma(x) = \max\{0, x\} = \frac{x + |x|}{2}, \quad x \in \mathbb{R}. \tag{24}$$

Thus by choosing one weight equal to 1 and one bias equal to $\gamma - 1$, the DNN can provide the exact solution (8) to this simple one-dimensional problem. This strongly suggests us choose ReLU as the activation function until further notice. To test the suitability of our choice, we explore how the activation function affects the learning results. The results in Fig. 3 clearly show that only ReLU captures an exact minimizer of the problem, whereas other choices including Tanh, Sigmoid, and even Leaky ReLU perform poorly. Note that the minimizer captured by ReLU is not (8), but any piecewise smooth function with slope 0 or 1 almost everywhere that satisfies the boundary conditions is also a (global) minimizer.

It is remarkable that the Deep Ritz method with ReLU can capture exact global minimizers in this simple problem due to the piecewise smooth character of this activation function. Other activation functions struggle and are unable to capture these weak solutions with derivative jumps. Later on, we propose a modification of ReLU to address more complicated problems, including regularized 1D and 2D curved interface problems.

3.1.2. Effect of the learning rate for 1D problem

We investigate how the learning rate affects the results for three values of the parameter $\gamma = 0.25, 0.5, 0.75$. We use the ReLU activation function. The results are shown in Fig. 4. We observe that if a large learning rate of 10^{-1} is used to train the loss function, the solution will converge to a constant $u(x) = \frac{\gamma}{2}$, which is a local minimum of the elastic energy but does not satisfy the boundary conditions. If a small learning rate is used, the solution will get stuck in another local minimum $u(x) = \gamma x$ for the case $\gamma = 0.25$, which is not a global minimum, although it satisfies the boundary conditions. For intermediate values of the learning rate, a global minimum is achieved. Also,

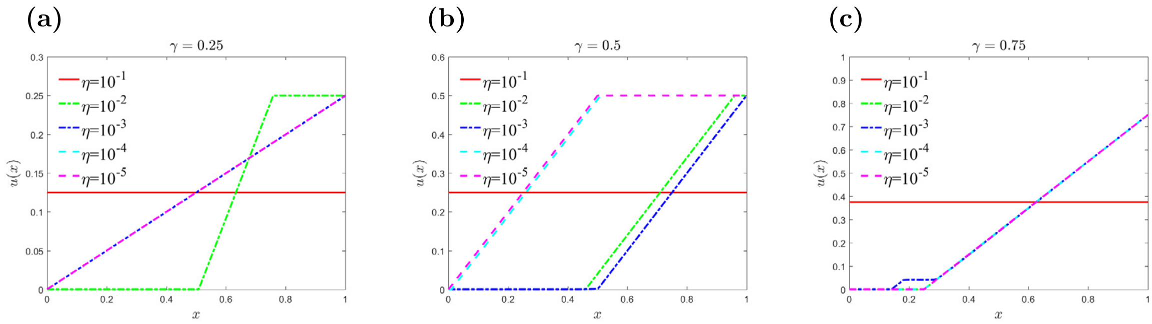

Fig. 4. 1D results with ReLU activation function and different learning rates with $iteration = 100,000$, NN: 3×128 (3 hidden layers and 128 neurons per layer). (a) $\gamma = 0.25$; (b) $\gamma = 0.5$; (c) $\gamma = 0.75$.

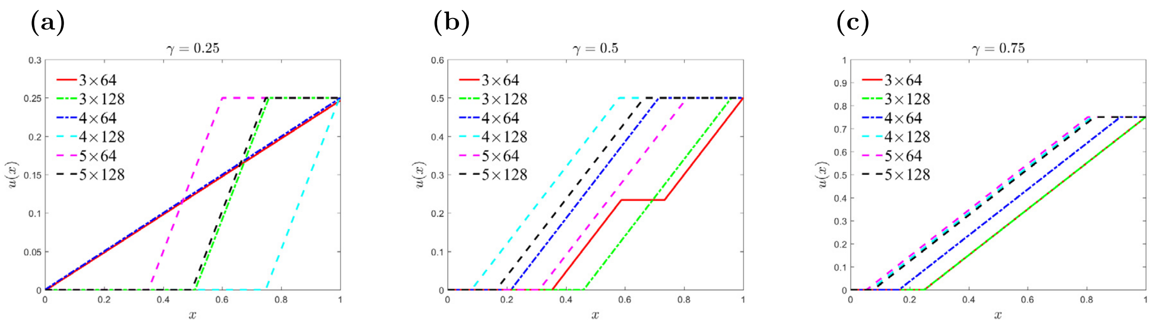

Fig. 5. 1D results with different size DNN. Here $iteration = 100,000$ and learning rate $\eta = 10^{-2}$. (a) $\gamma = 0.25$; (b) $\gamma = 0.5$; (c) $\gamma = 0.75$.

for these intermediate values of the learning rate, it is observed that different solutions, i.e., different combinations of line segments with slopes 0 and 1, are captured by different computations, which illustrates the loss of uniqueness of the minimization problem (6) as pointed out in Remark 2.1. Therefore, one should exercise caution in choosing the learning rate carefully to obtain the desired global minimum.

3.1.3. Effect of the DNN size for 1D problem

Next, we explore how the depth and width of the DNN affect the results when $\gamma = 0.25, 0.5, 0.75$. We use the ReLU activation function. The results are shown in Fig. 5. We find that for this 1D problem, different settings of depth and width can yield accurate results. For small values of γ, we need to use a larger DNN to approximate the solution u accurately. Again, we observe the evidence of loss of uniqueness of the minimization problem (6), i.e., different solutions are captured by different computations.

3.1.4. Study on the regularized 1D problem

In order to resolve the massive loss of uniqueness of minimizers for problem (20), it is common to add a higher gradient term to the energy; see Remarks 2.1 and 2.2 and [40]. Thus, we consider the following regularized minimization problem

$$\min E_\varepsilon\{u\}, \quad E_\varepsilon\{u\} = \int_0^1 \left[W(u'(x))dx + \frac{\varepsilon^2}{2}[u''(x)]^2 \right] dx, \tag{25}$$

where $\varepsilon > 0$ is a small parameter and boundary conditions are $u(0) = 0$ and $u(1) = \gamma$.

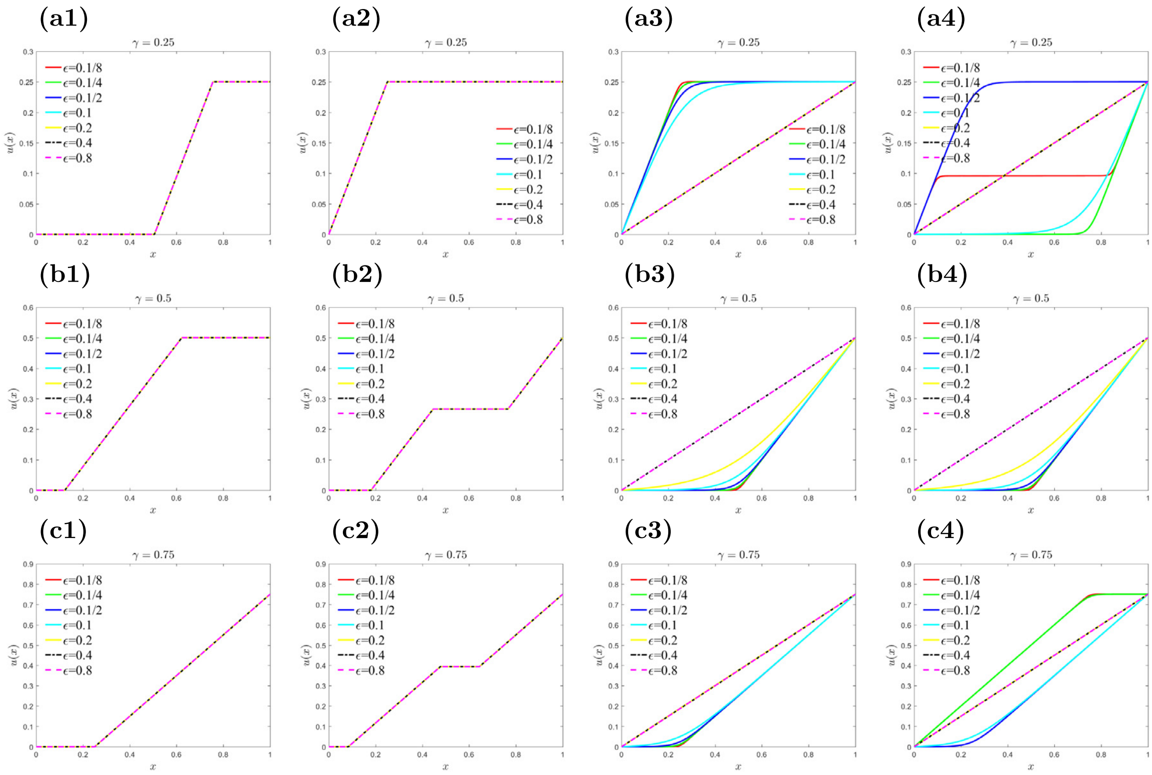

Fig. 6. 1D regularized problem results. Here $iteration = 100,000$, $\eta = 10^{-2}$ and NN: 3×128. (a) $\gamma = 0.25$; (b) $\gamma = 0.5$; (c) $\gamma = 0.75$. First column: Random initialization and ReLU activation function; Second column: $\gamma x + 0.1\sin(4x)$ initialization and ReLU activation function; Third column: Random initialization and SmReLU activation function; Last column: $\gamma x + 0.1\sin(4x)$ initialization and SmReLU activation function.

It is well known that the higher gradient term in the energy (25) smoothens derivative discontinuities, and replaces them with a smooth transition whose width is of order ε, e.g. [40]. Thus, we expect that the DNN with a ReLU activation function will not be able to capture this accurately due to its lack of smoothness. This motivates us to construct a new activation function by smoothening ReLU. In (24), we replace the absolute value $|x| = \sqrt{x^2}$ by $\sqrt{x^2 + \rho^2}$, where ρ is a small parameter. We call the resulting function a Smoothened Rectified Linear Unit or

$$\text{SmReLU:} \quad \sigma(x) = \frac{x + \sqrt{x^2 + \rho^2}}{2}, \quad x \in \mathbb{R}. \tag{26}$$

SmReLU is a smooth function for $\rho > 0$ and it reduces to ReLU for $\rho = 0$. We typically use $\rho = 0.1$ in the numerical computations of this paper.

Next, we consider how the initialization of u (i.e., the initial weights and biases of the DNN), the value of the higher gradient coefficient ε, and the activation function (ReLU vs SmReLU) affect our computations. The results are shown in Fig. 6. We use a 3×128 DNN with ReLU and SmReLU activation functions to approximate the solution u. The learning rate is $\eta = 10^{-2}$ and $\tau = 500$.

If we use the ReLU activation function, the optimal u is not differentiable at some points, and more than one derivative jump (kink) occurs when the initialization has oscillations; see Fig. 6. On the other hand, problem (25) has a unique smooth minimizer (modulo reflection about $x = 1/2$) [40] with a single smoothed kink (transition layer between slopes 0 and 1). This is well captured by the SmReLU DNN, which yields solutions with one smooth kink. The only exception, which has a local minimum with two kinks, occurs in Fig. 6 (a4). For larger values of ε, the DNN learns a smoother solution with a wider transition layer, in agreement with theory [40]. If ε is large enough, the regularization term dominates the loss function. So to minimize the second derivative energy term, u is very close to a linear function for this case. We conclude that the SmReLU activation function (26) works well for this problem and captures the expected behavior from the theory.

3.2. Numerical results for 2D problems

Our main goal in this work is to consider the following 2D nonconvex problem:

$$\min \int_{\Omega} W(\nabla u(x, y)) dx dy, \tag{27}$$

and its regularized counterpart (32). Here

$$W(\nabla u) = \frac{1}{2}\left(u_x^2(1-u_x)^2 + u_y^2\right), \tag{28}$$

and $\Omega = [0, L] \times [0, 1]$ with $L = 1$ or 2. We construct a fully connected DNN for approximating u with input (x, y). The loss function is defined as follows:

$$loss = loss_e + \tau loss_b, \tag{29}$$

where

$$loss_e = \frac{1}{N} \sum_{i=1}^{N} W\left(\nabla u_{NN}(x_i, y_i)\right),$$

$$loss_b = \frac{1}{N_b} \sum_{j=1}^{N_b} \left(u_{NN}(x_j^b, y_j^b) - u(x_j^b, y_j^b)\right)^2,$$

$\{(x_i, y_i)\}_{i=1}^{N}$ are the collocation points in the domain Ω, $\{(x_j^b, y_j^b)\}_{i=1}^{N_b}$ are the collocation points at the parts of the boundary where Dirichlet boundary conditions, namely the values of $u(x_j^b, y_j^b)$, are specified, while τ is the weight to scale and balance the losses from the energy ($loss_e$) and the boundary conditions ($loss_b$). In different settings, we also compute the energy using the following formula

$$E = h_x h_y \sum_{i=1}^{N_x} \sum_{j=1}^{Ny} W(\nabla u(x_i, y_j)),$$

where $h_x = h_y = 0.01$, $N_x = \frac{L}{h_x}$ and $N_y = \frac{1}{h_y}$.

3.2.1. Mixed boundary conditions

We first compute the problem (27) with the following mixed boundary conditions:

$$u(0, y) = 0, \ \ u(1, y) = \gamma, \quad 0 < \gamma < 1, \tag{30}$$

and the other two sides are free. In this case, the solution u is independent of y and minimizes the energy, if it is a solution of the 1D problem (20). We use both ReLU and SmReLU activation functions, and a 3×128 DNN to approximate u when $\gamma = 0.25$, 0.5, and 0.75.

Fig. 7 shows the results obtained by our method. In the top row of Fig. 7, we show results from the ReLU DNN, while the bottom row shows the SmReLU DNN minimizers. In the latter case, the solutions have only one gradient jump, whereas ReLU solutions typically have two. Our numerical results match the exact y-independent solutions of the 1D problem (6). It is interesting that the smoothened activation function SmReLU captures the sharp gradient jumps in this unregularized problem, as does ReLU, which was expected in view of the 1D results of Fig. 3. This occurs because SmReLU is rescaled by the weighs in the linear transformations T^i of the DNN in (11) to approximate sharp interfaces, but a little residual layer likely remains, that penalizes the energy to give only one jump. Here ReLU has a sharp gradient jump which does not contribute to the energy (6) so two jumps are observed.

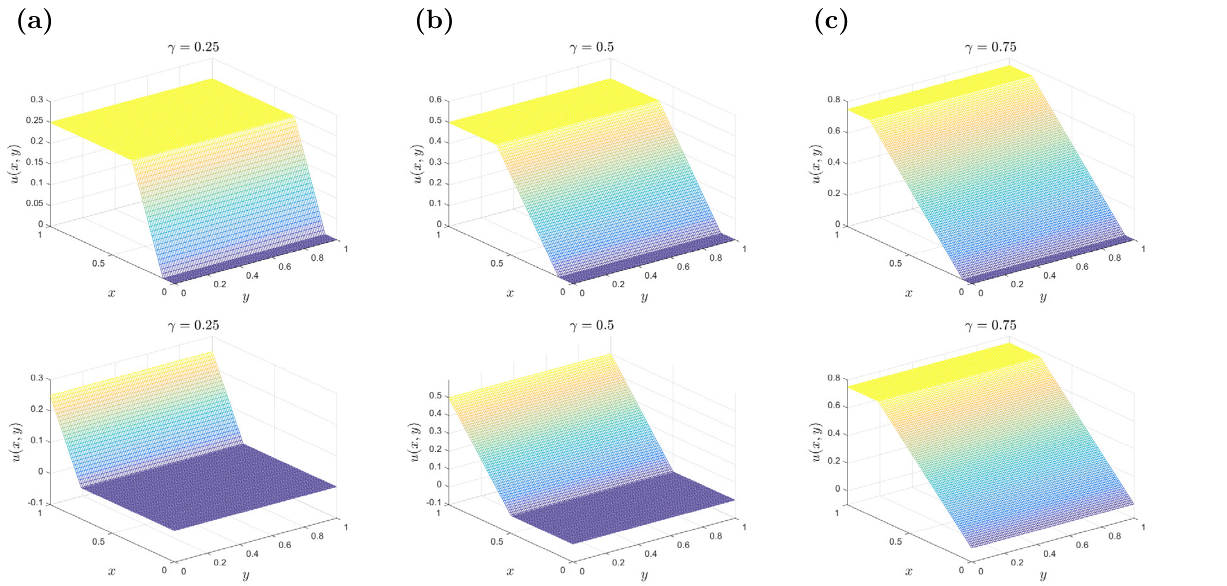

Fig. 7. 2D problem with a mixed boundary condition. (a) $\gamma = 0.25$; (b) $\gamma = 0.5$; (c) $\gamma = 0.75$. Top: ReLU activation function; bottom: SmReLU activation function.

3.2.2. Dirichlet boundary conditions

In the following, we will compute the problem (27) with Dirichlet boundary condition

$$u(x, y) = \gamma x, \quad 0 < \gamma < 1, \qquad \forall (x, y) \in \partial\Omega. \tag{31}$$

A typical value of interest is $\gamma = 1/2$. To achieve a zero-energy deformation, u_x should take values 0 or 1 with $u_y = 0$, which is what happens in Fig. 7 (the mixed boundary condition case considered in Section 3.2.1). On the other hand, the boundary condition (31) dictates that u_x is close to $1/2$ near the boundary, so there is an incompatibility between the energy wells and the boundary conditions. In fact, it is possible to show that problem (27) subject to (31) does not have a minimizer [16,34]. Instead, the energy can be effectively minimized in the limit by a sequence of deformations that involve a laminated microstructure, consisting of vertical parallel interfaces between domains with gradients in alternating wells $(u_x, u_y) = (0, 0)$ and $(1, 0)$. Such a minimizing sequence involves a larger and larger number of twin boundaries (jumps in ∇u along lines x =const.), effectively converging to a mixture of phases, where in the limit, u_x converges weakly to the average value $1/2$, making it compatible with the boundary conditions $u = x/2$ on $\partial\Omega$.

We show the results using the SmReLU activation function and $\gamma = 0.5$ in Figs. 8 and 9. Laminated microstructures prevail with alternating bands having u_x jumping between values near 0 (blue) and 1 (yellow) to minimize the energy density W, except near horizontal boundaries where these values are incompatible with boundary conditions (31) that require $u_x = 1/2$. This causes each band to split into two or more near the boundary as is also observed in experiments; see Fig. 1(a) [31].

In Fig. 8, we study the effect of the depth (number of layers) of DNN in capturing local minima with more and more bands. With only one layer, the solution is stuck at an unstable state (saddle point of W) with $u_x \approx 0.5$ in Fig. 8(b1). The number of bands increases as we increase the number of layers (depth) of the DNN, albeit with some oscillations. The energy also decreases and then appears to tend to a constant. In Fig. 9, we fix the depth as 5 and increase the width of the DNNs. The energy will decrease for wider DNNs. If we use larger DNNs (deeper or wider), we can get more refined microstructures in our results. However, we recall that there is no limit to the number and fineness of bands in this problem all the way to infinity [16,34], as there is no global minimum, but apparently a sequence of local minima with an increasing number of bands, like the ones observed here, that approach the infimum of energy. In a wider network, there will be more parameters (weights and biases) to be trained, rendering optimization more costly.

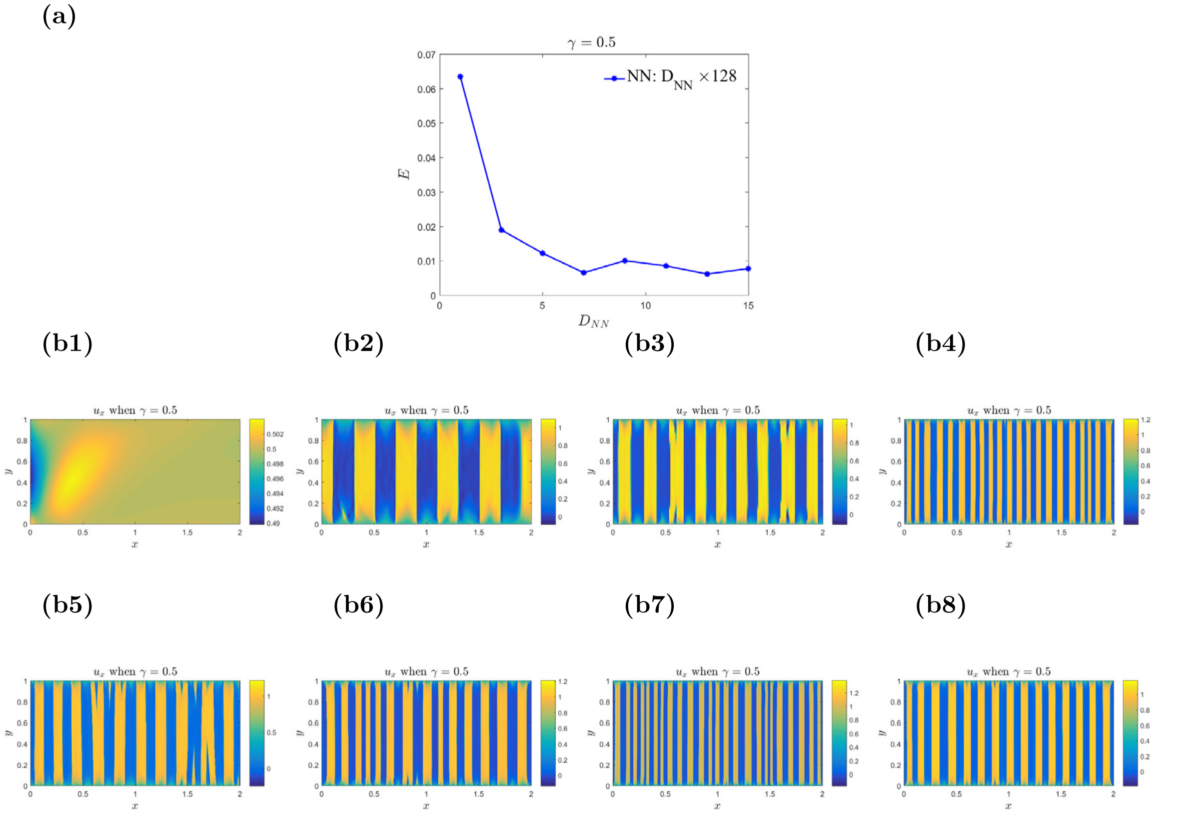

Fig. 8. 2D problem with Dirichlet boundary condition and SmReLU activation function: effect of varying depth. Here $\gamma = 0.5$, $iteration = 300,000$, $\eta = 10^{-3}$.(a) the energy E with different (b1) NN: 1 × 128; (b2) NN: 3 × 128; (b3) NN: 5 × 128; (b4) NN: 7 × 128; (b5) NN: 9 × 128; (b6) NN: 11 × 128; (b7) NN: 13 × 128; (b8) NN: 15 × 128.

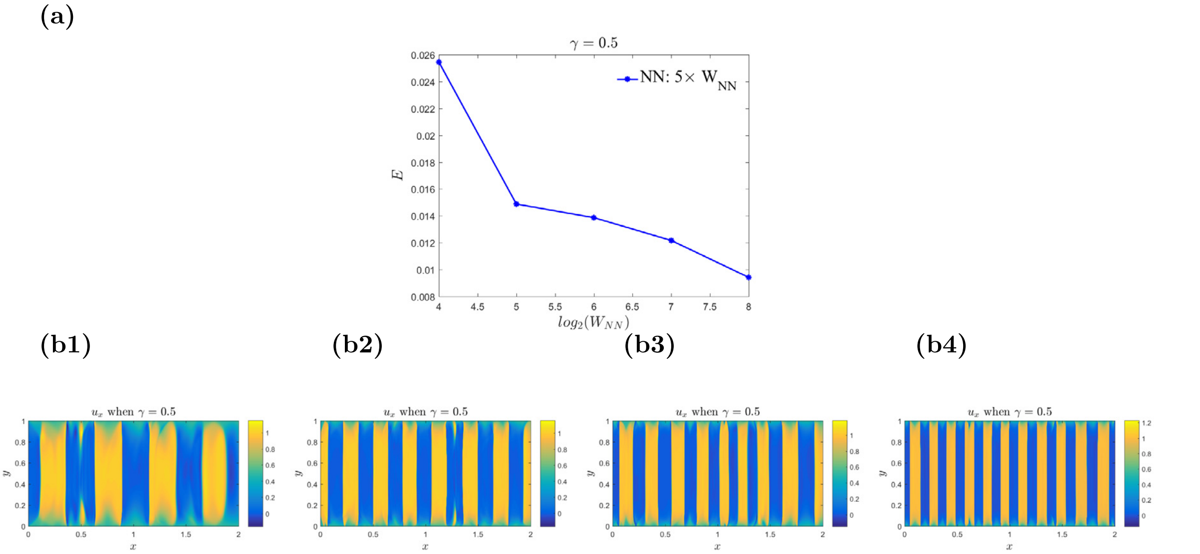

Fig. 9. 2D problem with Dirichlet boundary condition and SmReLU activation function: effect of varying width. Here $\gamma = 0.5$, $iteration = 300,000$, $\eta = 10^{-3}$. (a) the energy E with different (b1) NN: 5 × 16; (b2) NN: 5 × 32; (b3) NN: 5 × 64; (b4) NN: 5 × 256.

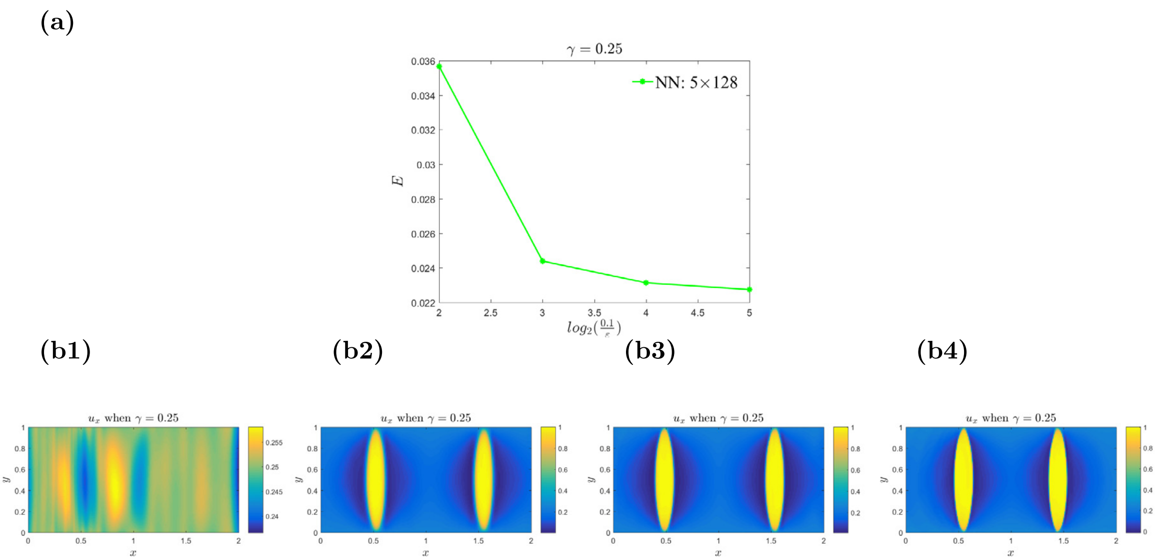

Fig. 10. 2D regularized problem with SmReLU activation function: effect of varying ε. Here $\gamma = 0.25$, iteration $= 300,000$, learning rate $\eta = 10^{-3}$ and NN: 5×128. (a) the energy with different regularized parameter $\varepsilon = \frac{0.1}{2^n}$ (b1) $n = 2$; (b2) $n = 3$; (b3) $n = 4$; (b4) $n = 5$.

3.2.3. Regularized scalar problem with Dirichlet boundary conditions

In order to study a problem that actually has a global minimum, we use the common approach of adding a higher gradient term and study the regularized problem as follows:

$$\min \int_{\Omega} \left[W(\nabla u(x, y)) + \frac{\varepsilon^2}{2} \left[\frac{\partial^2 u(x, y)}{\partial x^2} \right]^2 \right] dxdy, \tag{32}$$

where W is given by (28), $\Omega = [0, 2] \times [0, 1]$ and the boundary condition is the linear Dirichlet one (31). The higher gradient term renders local minima smooth and gradient discontinuities become smooth transition layers with the width of order ε [32,34,35,39,40]. Also, there is a global minimizer, but a careful bifurcation analysis [32] has revealed multitudes of stable local minima with multiple bands. A comparison of one of our solutions with one of the equilibria found in [32] is shown in Fig. 1(b) and Fig. 1(c).

We explore how the regularization parameter ε affects the energy. The results are shown in Figs. 10–12, where $\gamma = 0.25, 0.5, 0.75$ respectively. The energy decreases as the regularization parameter ε becomes smaller, while solutions for smaller ε have a finer microstructure with more bands, e.g., Fig. 11, as also observed in [32]. Fig. 13 shows the results of the DNN method with Tanh, ReLU and SmReLU activation functions. Again, we can see that the DNN with the SmReLU activation function clearly captures the laminated microstructures that prevail with alternating bands, while the DNN with Tanh and ReLU activation functions fails to do so.

3.2.4. Effect of the structures of the DNNs

In this subsection, we will explore more deeply how the structure of DNNs affects the results. Here, we fix the width of the DNN as 128 and $\varepsilon = \frac{1}{160}$. We explore how the depth of DNN affects the results, which are shown in Fig. 14. Once again, one layer is insufficient as it does not capture the banded microstructure. The energy decreases with the increasing depth of the DNN after some oscillations. The number of bands more or less increases with increasing depth, the maximum observed being 7 yellow bands. This is probably the solution with the most bands we can obtain by fixing the width at 128 and increasing the depth. More bands would increase the interfacial energy of transition layers, which is proportional to ε times the total length of transition layers between the blue and yellow phases. We also see a refinement mechanism in Fig. 11(b4), where a yellow band near the middle is split almost completely into two bands. The energy eventually plateaus out as the depth increases.

In order to explore how the width affects the results, we fix the depth of the DNN as 5 and $\varepsilon = \frac{1}{160}$. The results are shown in Fig. 15. We can see that the energy and the number of yellow bands decrease with increasing width

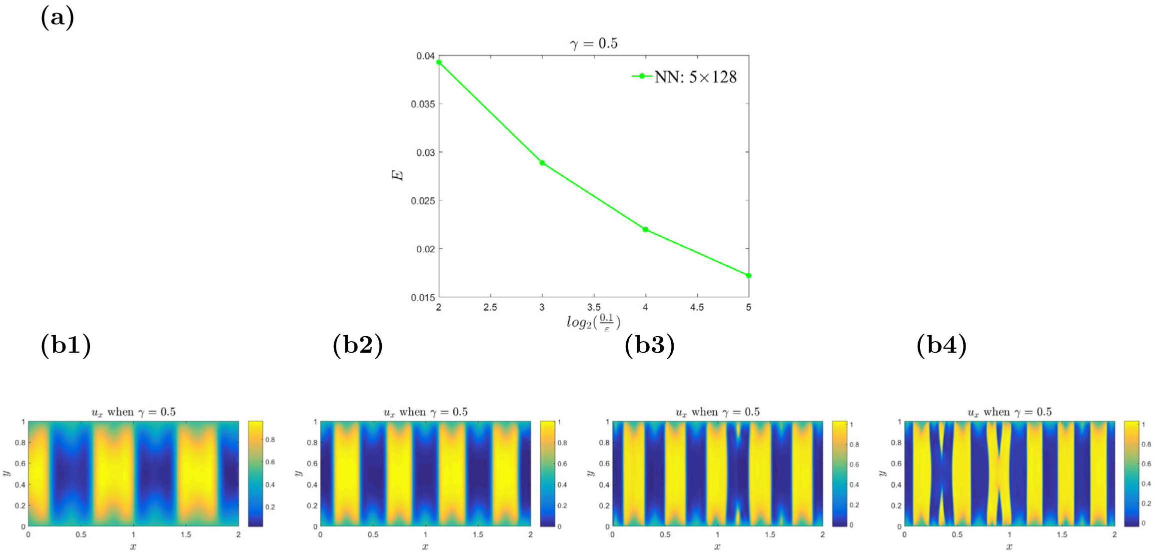

Fig. 11. 2D regularized problem with SmReLU activation function : effect of varying ε. Here $\gamma = 0.5$, iteration = 300, 000, learning rate $\eta = 10^{-3}$ and NN: 5 × 128. (a) the energy with different regularized parameter $\varepsilon = \frac{0.1}{2^n}$. (b1) $n = 2$; (b2) $n = 3$; (b3) $n = 4$; (b4) $n = 5$.

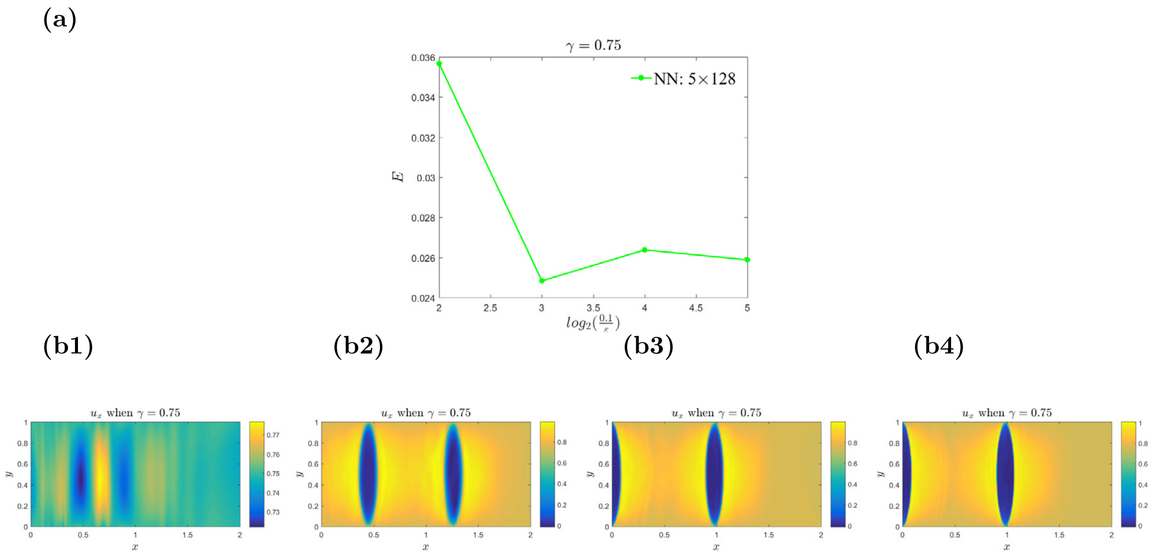

Fig. 12. 2D regularized problem with SmReLU activation: effect of varying ε. Here $\gamma = 0.75$, iteration = 300, 000, learning rate $\eta = 10^{-3}$ and NN: 5 × 128. (a) the energy with different regularized parameter $\varepsilon = \frac{0.1}{2^n}$. (b1) $n = 2$; (b2) $n = 3$; (b3) $n = 4$; (b4) $n = 5$.

of the DNN. For a fixed depth of 5 layers of the DNN, increasing the width beyond 128 does not result in more than 5 yellow bands in our solutions. The results in Figs. 14 and 15 indicate that depths play a more important role.

3.2.5. Distribution of the collocation points

In the numerical result of minimizers, we observe that the energy density near the boundaries is higher, whereas we use a random choice of collocation points in the domain Ω. In order to minimize the energy contribution near the boundary more effectively, we select a higher proportion of collocation points near the top and bottom parts of the domain. We divide the domain Ω into three parts, $\Omega_1 = (0, 2) \times (0, 0.15)$, $\Omega_2 = (0, 2) \times (0.15, 0.85)$ and

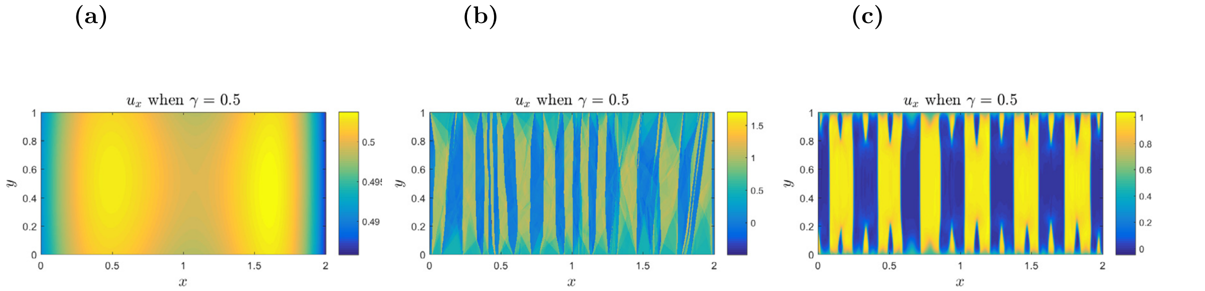

Fig. 13. Different activation functions in the 2D regularized problem. Here iteration = 300, 000, $\eta = 10^{-3}$ and $\varepsilon = \frac{0.1}{32}$. (a)Tanh activation function; (b) ReLU activation function; (c) SmReLU activation function.

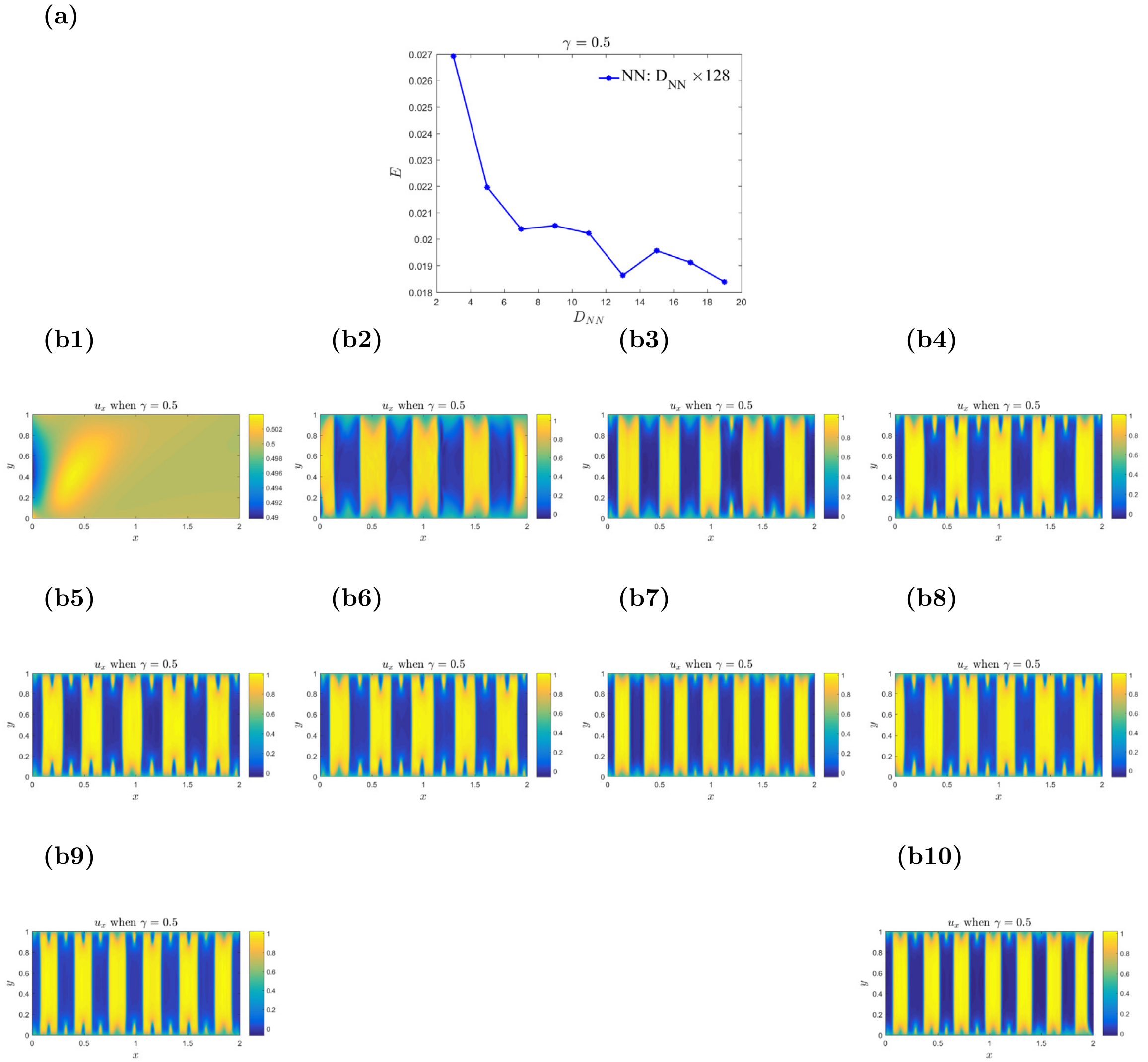

Fig. 14. 2D regularized problem with SmReLU activation function. Here $\gamma = 0.5$, $iteration = 300,000$, $\eta = 10^{-3}$ and $\varepsilon = \frac{0.1}{16}$. (a) the energy with different (b1) NN: 1 × 128; (b2) NN: 3 × 128; (b3) NN: 5 × 128; (b4) NN: 7 × 128; (b5) NN: 9 × 128; (b6) NN: 11 × 128; (b7) NN: 13 × 128; (b8) NN: 15 × 128; (b9) NN: 17 × 128; (b10) NN: 19 × 128.

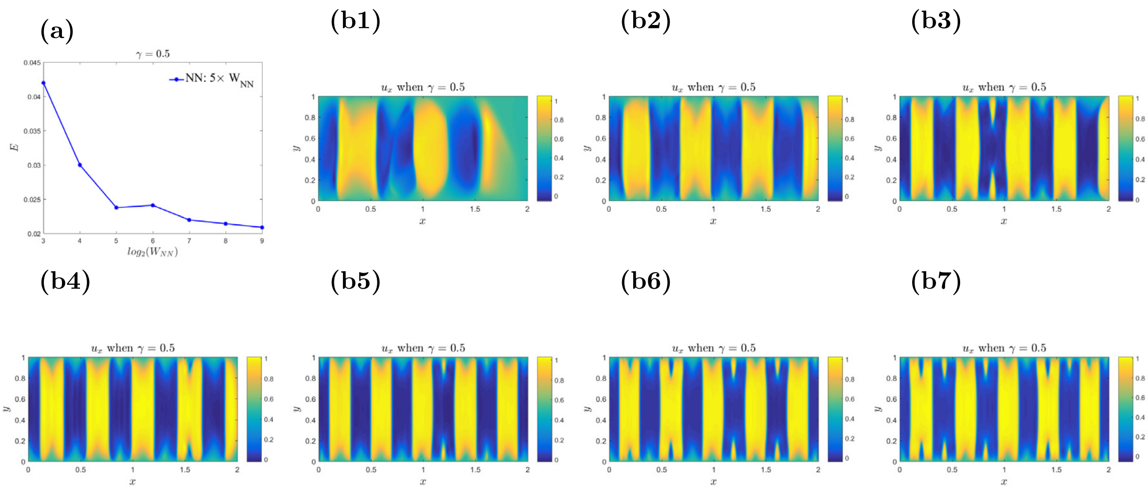

Fig. 15. 2D regularized problem with SmReLU activation function. Here $\gamma = 0.5$, $iteration = 300,000$, $\eta = 10^{-3}$ and $\varepsilon = \frac{0.1}{16}$. (a) the energy with different (b1) NN: 5×8; (b2) NN: 5×16; (b3) NN: 5×32; (b4) NN: 5×64; (b5) NN: 5×128; (b6) NN: 5×256; (b7) NN: 5×512.

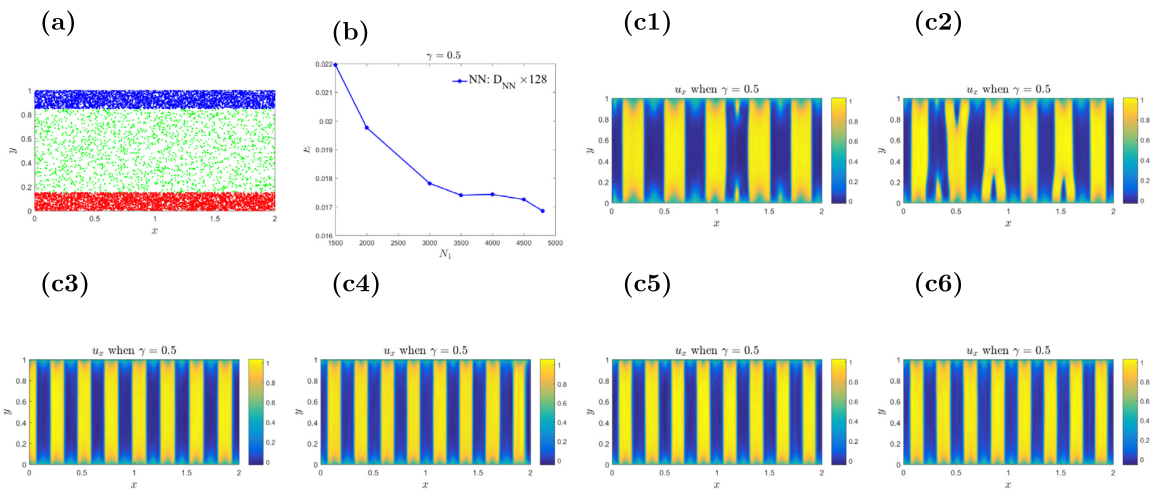

Fig. 16. Adapted collocation points in the 2D regularized problem with SmReLU activation function, where $\gamma = 0.5$, $iteration = 300,000$, $\eta = 10^{-3}$ and $\varepsilon = \frac{0.1}{16}$. (a) the position of collocation point; (b) the energy with different (c1) $N_1 = N_3 = 1500$ and $N_2 = 7000$; (c2) $N_1 = N_3 = 2000$ and $N_2 = 6000$; (c3) $N_1 = N_3 = 3000$ and $N_2 = 4000$; (c4) $N_1 = N_3 = 3500$ and $N_2 = 3000$; (c5) $N_1 = N_3 = 4000$ and $N_2 = 2000$; (c6) $N_1 = N_3 = 4500$ and $N_2 = 1000$.

$\Omega_3 = (0, 2) \times (0.85, 1)$. In each domain, we randomly choose N_1, N_2, and N_3 points, and let $N_1 = N_3$. For details see Fig. 16. When we increase the number $N_1 = N_3$, keeping the sum of N_1, N_2, and N_3 fixed, the energy decreases.

3.3. Numerical results for 2D problems with curved interfaces

To further investigate the performance of our method, we consider the following energy

$$W(\nabla u) = \frac{1}{2}\Big((\cos(\phi)u_x + \sin(\phi)u_y)^2(1 - (\cos(\phi)u_x + \sin(\phi)u_y))^2 + (-\sin(\phi)u_x + \cos(\phi)u_y)^2\Big), \tag{33}$$

with a Dirichlet boundary condition

$$u(x, y) = \gamma(\cos(\phi)x + \sin(\phi)y), \quad \forall (x, y) \in \partial\Omega. \tag{34}$$

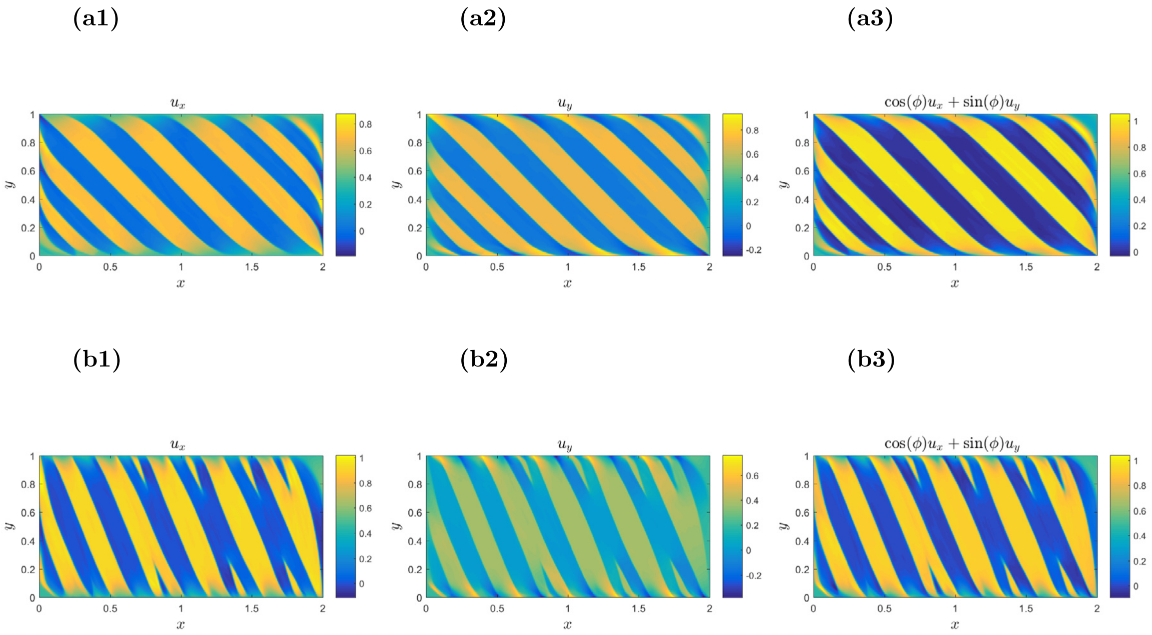

Fig. 17. Rotated minimum 2D problem with SmReLU activation function. Here $iteration = 300,000$, $\eta = 10^{-3}$ and $\varepsilon = \frac{0.1}{32}$. (a) $\gamma = 0.5$ and $\phi = \frac{\pi}{4}$; (b) $\gamma = 0.5$ and $\phi = \frac{\pi}{8}$. left: u_x; middle: u_y; right: $\cos(\phi)u_x + \sin(\phi)u_y$.

This experiment corresponds to rotating one of the minima of W from $(0, 1)$ to $(\cos\phi, \sin\phi)$, where ϕ is a fixed parameter. The results are shown in Fig. 17. We find that the incompatibility with the boundary conditions now forces the phase boundaries (interfaces) to bend and curve, demonstrating the advantage of the mesh-free nature of the DNN method. Therefore, our method is advantageous in dealing with phase transition problems with complex geometry.

4. Conclusions and discussion

This work makes use of the Deep Ritz Method for the first time in order to solve a challenging nonconvex gradient-energy minimization problem from materials science. Such problems are difficult for two reasons. Firstly, they typically do not possess a global minimum, but only minimizing sequences and a complex set of local minima. Secondly, minimizers, including local ones, are characterized by complicated geometric microstructures that comprise multiple layers separated by phase interfaces (twin boundaries). These interfaces act as free boundaries and are characterized by gradient discontinuities. Additionally, these interfaces exhibit needle tapering and tip-splitting topological transitions near the boundaries of the domain [32,34–36,38,39].

In our approach, the Deep Ritz method is used to represent the candidate minimizer as a deep neural network, allowing the energy to be expressed as a function of the weights and biases (parameters in the DNN), and optimization to be performed with respect to them. Our results demonstrate that the Deep Ritz method is capable of spontaneously capturing the complexities of local or global minimizers, including the microstructure complexities arising from gradient-energy minimization problems. A key new ingredient of our method is the activation function that we propose for the first time. We begin by observing that the ReLU activation function can capture exact global minimizers in a simple unregularized 1D problem due to its piecewise linear nature. In contrast, other activation functions such as Tanh and Sigmoid struggle and are unable to approximate these weak solutions with derivative jumps (Fig. 3). For regularized problems, we introduce a new, smoothened version of the ReLU activation function, which we call SmReLU (see Eq. (26)). This activation function is successful in producing results that agree with careful finite difference computations [32] and is able to capture the correct microstructure in Fig. 1(b) and Fig. 1(c). In comparison, other traditional activation functions fail to capture the same microstructure (Fig. 13).

Our findings suggest that both width and depth are essential for the DNN to capture local minimizers accurately. In general, increasing the number of layers (depth) increases the number of twin bands appearing in our solution until a point where further depth increase does not improve the solution. In that sense, depth plays a role analogous to mesh size in the FEM. Additionally, we find that for a fixed depth, it is not necessary to increase width past a certain point, making depth more important in this case.

Equipped with our SmReLU activation function, our method is capable of capturing both sharp gradient discontinuities across phase/twin boundaries and smoothed transition layers. Moreover, our method can approximate curved interfaces successfully, as shown in Fig. 17. The mesh-free nature of our approach is a significant advantage over the FEM and FDM, particularly in the presence of interfacial discontinuities carrying surface energy, where issues can arise due to mesh misorientation compared to interfaces [46]. Another advantage of our method is its simplicity, ease of programming, and ability to naturally capture nonsmooth solutions with geometrically complex free interfaces in more than one dimension without special treatment, such as discontinuous Galerkin FEM for higher gradients in nonconvex elasticity [47].

CRediT authorship contribution statement

Xiaoli Chen: Conceptualization, Programming, Methodology, Writing – original draft, Writing – review & editing. **Phoebus Rosakis:** Conceptualization, Methodology, Writing – review & editing. **Zhizhang Wu:** Conceptualization, Programming, Methodology, Writing – original draft, Writing – review & editing. **Zhiwen Zhang:** Conceptualization, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The research of X. Chen is supported by the Ministry of Education, Singapore, under its Research Centre of Excellence award to the Institute for Functional Intelligent Materials (I-FIM, project No. EDUNC-33-18-279-V12). The research of Z. Zhang is supported by Hong Kong RGC grant (projects 17300318 and 17307921), National Natural Science Foundation of China (project 12171406), the outstanding young researcher award of HKU, Hong Kong (2020–21), Seed Funding for Strategic Interdisciplinary Research Scheme 2021/22 (HKU, Hong Kong), and an R&D Funding Scheme from the HKU-SCF FinTech Academy.

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